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Advanced Topics in Software Engineering

Set 2: Machine Learning Foundation

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What is Generative AI?

- A subset of Artificial Intelligence focused on generating new data from existing data used for training and from context provided to it
  - Training datasets may consist of text (words), images (pixels), and speech (signals)
- Main goal: Predict probability of the word following an incomplete (masked) sentence
- Buyers beware:
  - Possibility of creating fake content that looks legitimate (e.g., fake news, fake pictures, etc.)
  - Possible loss of privacy (Do not feed private info into LLM!)
  - Bias in the answers (reflecting bias in the training data)
  - Job displacement
  - Environmental impact

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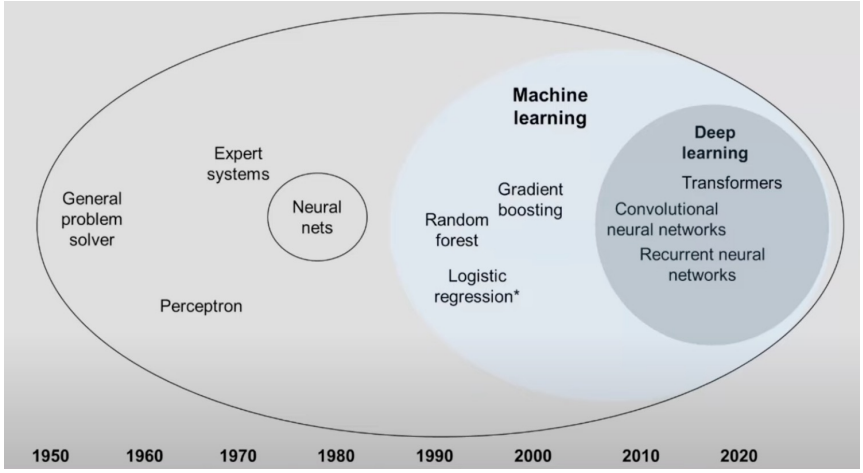
Historical perspective

- AI has existed since the 1950's
- Evolutionary stages (gross simplification)
  1. Algorithmic AI

     Robotics, Vision, NLP, Expert Systems, etc.
  2. Predictive AI

     Regression, SVMs, NNs
  3. Generative AI

     LLMs



Source: Jesse Spencer Smith

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The beginnings (1950's-80's)

- GPS (General Problem Solver) 1957 – A universal problem-solving machine using predicate logic
  - Success: Towers of Hanoi
  - Failure: Everything else (combinatorial explosion of search space)
- Better luck with expert systems
  - Rule-based
  - MYCIN: Medical diagnosis and treatment recommendations based on symptoms and test results
  - Tied or beat human doctors at diagnosing blood diseases
- Also, research in vision, natural language processing, speech recognition, robotics planning, etc.
- Big promises: Few goods delivered
- The AI winter followed

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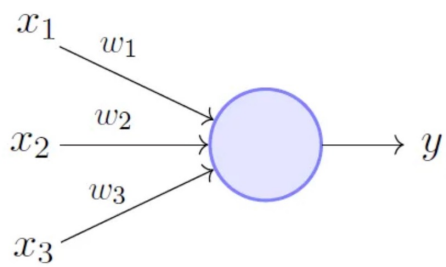
Perceptron

- Perceptron = Single artificial neuron
- Invented at UIC by McCulloch and Pitts in 1943
  - Neural nets were born at UIC!
- Reprised by Minsky and Papert in late 1960's
- Emulate behavior of neurons in human brain
- Given a set of inputs x_i and weights w_i, produce output y
- Neural nets are made of perceptrons, usually arranged in layers
- Perceptrons in a layer receive inputs from the previous layer and feed into the next layer



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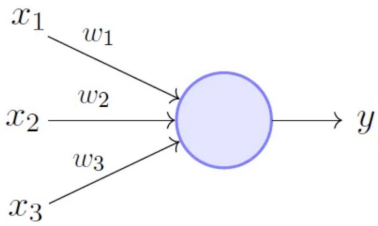
Perceptron vs. human neuron



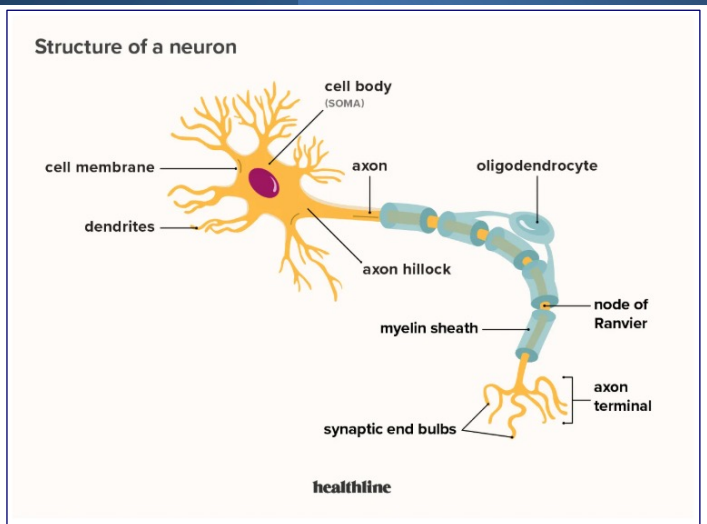


Illustration by Sophia Smith

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Next evolution stage statistical learning (1990's)

- Multiple kinds of models: Regression, Logistic Regression, Random Forests, etc.
- Supervised (data is labeled with ground truth) vs. unsupervised learning (no ground truth)
- Train model to solve a given problem
- Predictive AI: *Regression* vs. *Classification* problems
- AI gets more popular
- The onset of *Deep Learning*
  - Built upon previous work on neural nets
  - Bigger, more powerful nets with multiple hidden layers
  - Specialized nets (Convolutional Neural Nets and Recurrent Neural Nets)
  - Able to handle unstructured data (no need for fancy knowledge representation algorithms)

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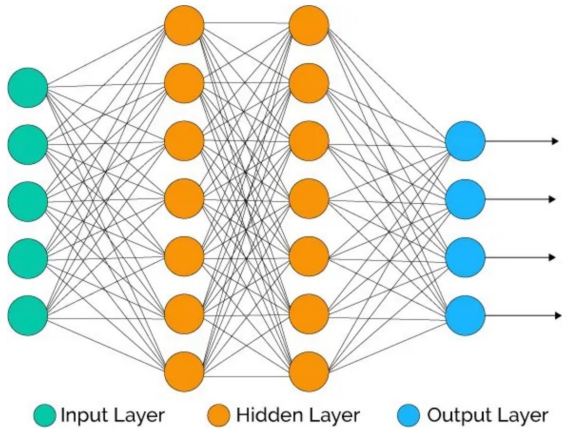
A very small neural net

- Typical structure: 1 Input layer, 1 output layer, *n* hidden layers in between
- Each layer is made of perceptrons (cells)
- Supervised learning
  - Given training data set containing *x* data points
  - Each data point contains (input, output) pair
  - Data points are fed into net
  - At each run, weights on the cells are changed to reward cells that "guessed right" and punish cells that led to wrong guess
  - Big advantage: Handle unstructured data



Source: medium.com/Shaistha Fathima

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Current evolution stage: Generative AI

- Generative AI: Able to create new and original content
- Enabled by computing power available today
- Transformer architecture
- *Self-supervised learning*: Guess missing words from sentences (Masked Words)
  - No need to label data as with old supervised learning
  - Deep model trained by *backpropagation*
- Trained model can put together full sentences
- Training corpus: books, articles, blogs, etc. publicly available on the Internet
  - Is everything in the Internet accurate?
- Most models cannot be retrained
  - To improve answers, provide more context; Prompts provide context

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Transformer architecture

- Introduced by Google Brain group in 2017 (https://research.google.com/teams/brain/)
  - Google Brain also created TensorFlow ML library in 2017
- Architecture of Large Language Models
- Typically, two main components
  1. Encoder = Understand user input
  2. Decoder = Create AI output
- Encoder produces *embeddings* – Numerical vectors that carry semantic information based on word similarity in a so-called "latent space"
  - Words appearing in similar contexts will have similar numerical vectors
  - E.g., words "monarch" and "queen" will have similar but not identical numerical vectors
  - Vectors are added together for meaning (e.g. Queen == Monarch + Woman == King – Man + Woman)

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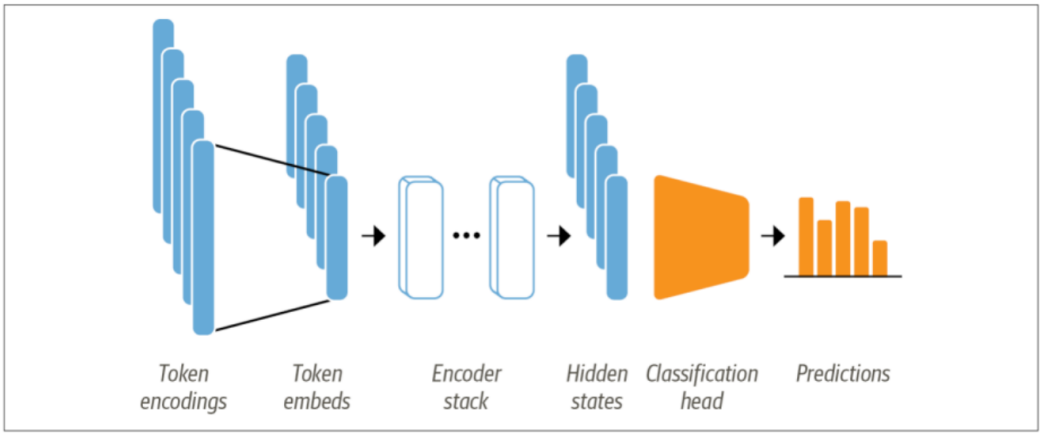
Structure of ChatGPT: The transformer architecture



Source: L. Turnstall, L. von Werra, and T. Wolf: Natural Language Processing with Transformers

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The structure explained

- Token encodings: Words or parts of words
  - Vocabulary typically consists of 100,000s of words
- Tokens (aka words or parts of words) are converted to numerical vectors in a low dimensional vector space
  - Each token is represented by a vector; vectors are created by a NN with supervised task
  - Goal: Similar tokens mapped to similar vectors (with similar values in matching positions) in vector space
- Encoder stack = NN layers encoding knowledge learned by harvesting Internet text
  - Frozen in time (not updated in real-time)
  - Weights cannot be changed
- Classifier head: Shallow net (trainable) using hidden state output by encoder stack to predict masked word

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Transformer architecture

- Introduced by Google Brain group in 2017 (https://research.google.com/teams/brain/)
  - Google Brain also created TensorFlow ML library in 2017
- Architecture of Large Language Models
  - General LLMs: GPT (Generalized Pretrained Transformer – OpenAI), Bert (Bidirectional Encoder Representation from Transformers – Google), Llama (Large LAnguage model Meta Ai – Meta)
  - Task-specific LLMs: Code Llama (Meta), Claude Code (Anthropic)
  - Domain-specific LLMs: E.g., BloombergGPT for finance domain (https://arxiv.org/abs/2303.17564), CaseHOLD for Law (https://arxiv.org/pdf/2104.08671), Med-PaLM for medical domain (https://sites.research.google/gr/med-palm/)
- Distinguish NL interface (e.g., ChatGPT, Bard) from underlying LLM (e.g., GPT, Bert)

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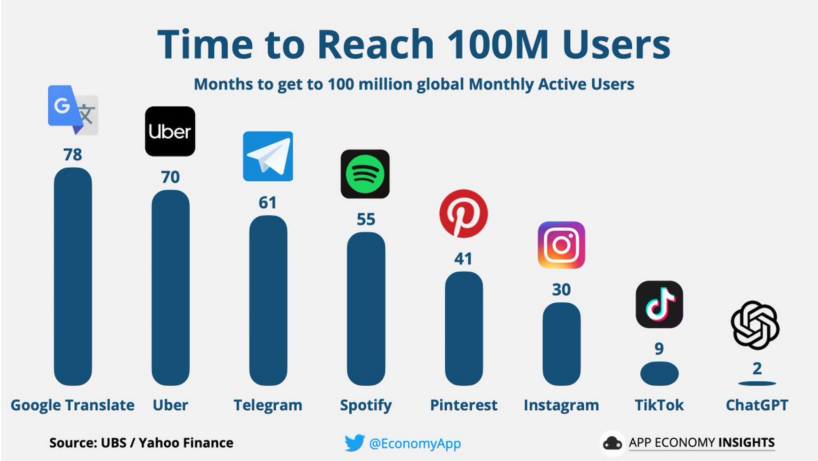
GPT-3 vs. GPT-4

- GPT-3: 570 GB of compressed text data, 175 Billion parameters
- GPT-4: TBs of text data, 1,800 Billion parameters, improved functionality to GPT-3
- ChatGPT reached 100 Million users faster than all other popular web apps



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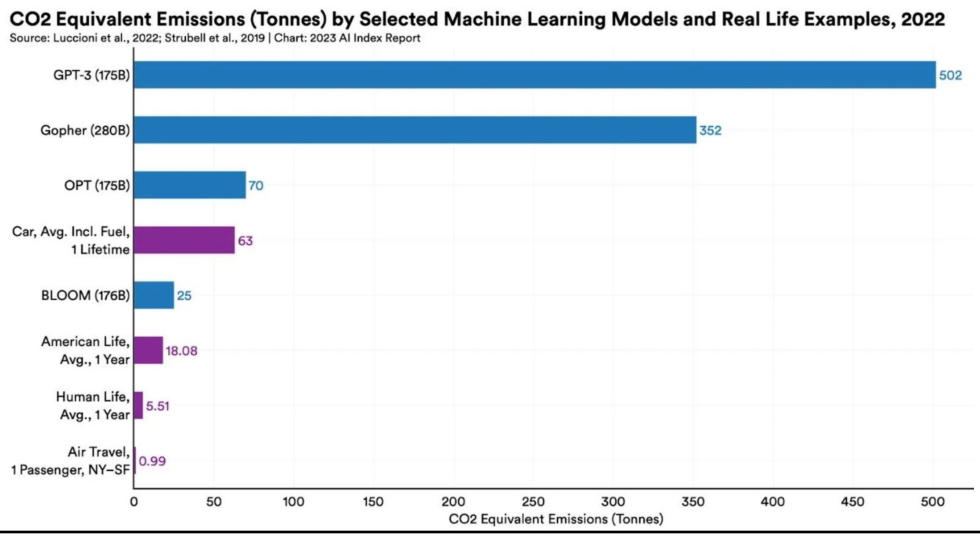
Environmental effects (Source: Mack DeGeurin, Gizmodo)



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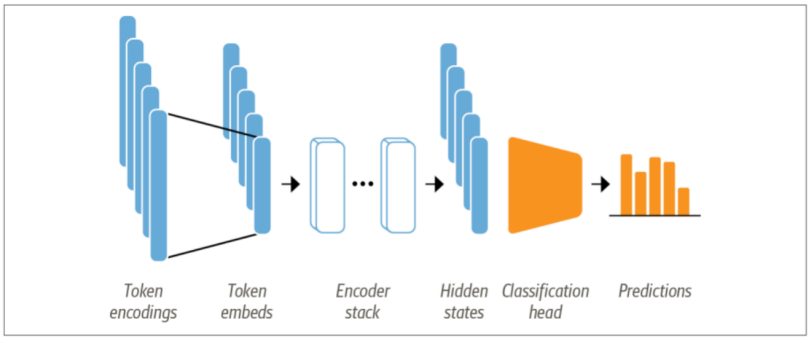
Training ChatGPT

- Huge corpus of text from the Internet
- Ask to guess missing word from a sentence
- If wrong word guessed, adjust weights by back propagation (similar to tuning vanilla NN)



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Popular LLMs

LLM	Developer	Multimodal?	Reasoning?	Access
GPT-5	OpenAI	Yes	Yes	API, Chatbot
gpt-oss	OpenAI	No	Yes	Open
GPT-4o	OpenAI	Yes	No	API, Chatbot
o3 / o3-mini	OpenAI	No	Yes	API
Gemini	Google	Yes	No	API, Chatbot
Gemma	Google	No	No	Open
Llama	Meta	Yes	No	Open
R1	DeepSeek	No	Yes	Open, API, Chatbot
V3.1	DeepSeek	No	Yes	Open, API, Chatbot
Claude	Anthropic	Yes	Yes	API, Chatbot

Source: https://zapier.com/blog/best-llm/

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Criteria for choosing an LLM

1. Task alignment
   - Choose an LLM appropriate for your task
   - Claude Code or Code Llama for software engineering work
2. Model size
   - Run locally vs. the cloud
3. Training data
4. Support available?

Useful sites

- Huggingface
  - Repositor of LLMs, datasets and apps
- LM Studio
  - Support for running LLMs locally
- Ollama
  - LLM repository

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Setting up Claude

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Installing and setting up Claude Code

- On Mac: Use command curl -fsSL https://claude.ai/install.sh | bash
  - Choose environment of use (terminal, browser, etc.)
- On Windows: Use Powershell command curl -fsSL https://claude.ai/install.sh | bash
- Using NodeJS
  1. Install NodeJS: Go to https://nodejs.org/en and install NodeJS for Windows
  2. Use *npm* to install Claude-code npm install –g @anthropic-ai/claude-code
- After installing, enter claude at command prompt to set up claude-code
- Claude code site: https://claude.com/product/claude-code
- Excellent video by Kevin Stratvert https://www.youtube.com/watch?v=eMZmDH3T2bY.

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